

Reverse Linked List

Pattern:

- Linked List
- Two Pointers
- Pointer Manipulation

How to recognize

- Question asks you to reverse a linked list
- Need to change node connections, not values

Interview Approach:

```
prev = null
```

```
curr = head
```

```
while curr:
```

```
    nextNode = curr.next
```

```
    curr.next = prev
```

```
    prev = curr
```

```
    curr = nextNode
```

```
return prev
```

Key Idea:

- Always save the next node before changing pointers

Contains Duplicate

Pattern:

- Hash Table
- Array

How to recognize:

- Looking for duplicates
- Needs fast lookup

Interview Approach:

Create empty set

For each number:

 If number in set:

 return True

 Add number to set

return False

Key Idea:

- Set provides $O(1)$ lookup

Remove Linked List Elements

Pattern:

- Linked List
- Dummy Node

How to recognize:

- Delete nodes with a specific value
- Head node might need removal

Interview Approach:

- dummy -> head

```
curr = dummy
```

```
while curr.next:
```

```
    if curr.next.val == target:
```

```
        curr.next = curr.next.next
```

```
    else:
```

```
        curr = curr.next
```

```
return dummy.next
```

Key Idea:

- Use a dummy node